

Arash Asgari

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SUMMARY

LLM/VLM Research Engineer and Applied ML Researcher with 3+ years of experience in foundation models, large-scale model evaluation, agentic bias mitigation, fine-tuning, data curation, and efficient ML systems. Strong hands-on expertise in PyTorch, Hugging Face, and Transformers, with experience building reproducible multimodal benchmarking pipelines, optimizing model inference, and developing scalable research prototypes. First author of an ACL 2026 paper, with research spanning LLM and VLM evaluation, representation engineering, knowledge editing, and parameter-efficient fine-tuning.

WORK EXPERIENCE

Machine Learning Researcher, York University

May 2024 - August 2026

- **Designed and implemented a large-scale evaluation pipeline** to benchmark AI fairness and bias, devising novel statistical agreement metrics to cross-validate performance across 8 open-source LLMs (ranging from 2B to 9B parameters) and 7 distinct evaluation frameworks (this work is published at ACL 2026).
- **Co-developed ChartBias and a multi-agent bias-mitigation framework** for vision-language chart interpretation, evaluating 12 VLMs on 155,484 responses across 820 real-world charts and using chart-grounded evidence extraction with a counterfactual judge to reduce narrative shift (accepted to EMNLP 2026).
- **Engineered advanced inference-time and training-time debiasing algorithms** using representation engineering, successfully leveraging persona vectors and activation space steering to mitigate model bias (submitted to EMNLP).
- **Executed rigorous adversarial testing and safety benchmarking** for state-of-the-art LLMs in medical QA, utilizing baseline, jailbreak, and in-context debiasing prompts to evaluate model resistance and vulnerability (this work is published in AAAI symposium).

Research Intern, Vector Institute

January 2025 - May 2025

- **Engineered a comprehensive evaluation pipeline** to benchmark state-of-the-art LLM knowledge editing methods (e.g., ROME, MEMIT, and MEND), working under the supervision of Dr. Frank Rudzicz.
- **Deployed and tested large language models** on high-performance compute clusters, optimizing scripts for memory-efficient inference and data processing.
- **Analyzed model performance and robustness** using medical QA datasets like MedSynth, evaluating the specificity and generalization of post-edit model weights.

Software Engineering Researcher, Sharif University of Technology

October 2021 - April 2024

- **Curated and open-sourced DistilKaggle**, a massive-scale benchmark dataset comprising over 542,000 Jupyter notebooks, engineering an extraction pipeline to calculate 34 distinct structural and script-based code metrics (this work is published in MSR conference).
- **Engineered an automated code evaluation framework** by fine-tuning a DistilBERT transformer for semantic opinion mining, devising a novel metric that classified the code quality of 132,000+ computational notebooks with 89% accuracy (this work is published in Empirical Software Engineering Journal).
- Conducted rigorous **performance benchmarking of ML inference pipelines** using NVIDIA Multi-Instance GPU (MIG) architecture, analyzing hardware utilization, throughput, and latency across varying load profiles.

Machine Learning Engineer, Drivee, California, US

February 2023 - January 2024

- **Architected a serverless image processing pipeline** using Google Cloud Functions in a pipe-and-filter architecture, sequentially automating exposure regularization, vehicle smoothing, and image enhancements.
- **Optimized a YOLO-based background removal model** with quantization and ONNX for memory-efficient deployment on Google Cloud Functions, powering the Drivee iOS app's 360° vehicle imaging pipeline.
- **Developed an autonomous LLM agent** using LangChain and OpenAI tool calling, integrating Google Calendar and live dealership databases for real-time customer assistance.
- **Built a voice-based LLM agent** using streaming audio and response generation to reduce latency and enable real-time conversational interactions.

EDUCATION

2024 - Aug 2026	M.Sc (Computer Science) at York University	(GPA: 4.0/4.0)
2021 - 2024	M.Sc (Computer Engineering) at Sharif University of Technology	(GPA: 18.28/20.0)
2017 - 2021	B.Sc (Computer Engineering) at K.N.T.U University of Technology	(GPA: 19.57/20.0)

SKILLS

LLM & Agentic AI	PyTorch, Hugging Face, Transformers, LangChain, MCP, ADK (Agent Development Kit), Tool/Function Calling, Prompt Engineering, RAG, LLM Fine-tuning
ML & Model Optimization	TensorFlow, Scikit-learn, ONNX, Model Quantization, Efficient Inference
MLOps & Cloud	Docker, Kubernetes, AWS (S3, EC2, Redshift), CI/CD, Model Monitoring, W&B, FastAPI, Linux, Bash
Data Platforms	Spark (PySpark, SparkML), Hadoop (MapReduce), Hive, SLURM
Languages	Python, JavaScript, SQL, R
Data Analysis & Visualization	Pandas, NumPy, Matplotlib, Seaborn
Databases	MySQL, PostgreSQL, MongoDB

PUBLICATIONS

Arash Asgari et al. (2026). “Quantifying Metric and Model Agreement in Bias Evaluation of Large Language Models”. In: *The 64th Annual Meeting of the Association for Computational Linguistics*.

Asgari, Arash et al. (2026). “Fairness Without Forgetting: Improving LLM Bias via Persona Vectors”. In: *Submitted to the 2026 Conference on Empirical Methods in Natural Language Processing*. Under review.

Chen, Chi-Yu et al. (2026). “Algorithms Trained on Normal Chest X-rays Can Predict Health Insurance Types”. In: *Proceedings of the 9th International Conference on Medical Imaging with Deep Learning*. Vol. 315. Proceedings of Machine Learning Research. PMLR, pp. 4166–4181. URL: <https://proceedings.mlr.press/v315/chen26c.html>.

Mizanur Rahman, Huan Wu, Arash Asgari et al. (2026). “Same Chart, Different Story: Bias in Vision-Language Chart Interpretation”. In: *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*. Accepted.

Asgari, Arash et al. (Nov. 2025). “MedPerturbing LLMs: A Comparative Study of Toxicity, Prompt Tuning, and Jailbreaks in Medical QA”. In: *Proceedings of the AAAI Symposium Series* 7.1, pp. 438–447. DOI: [10.1609/aaais.v7i1.36916](https://doi.org/10.1609/aaais.v7i1.36916). URL: <https://ojs.aaai.org/index.php/AAAI-SS/article/view/36916>.

Mojtaba Mostafavi Ghahfarokhi, Alireza Asadi, Arash Asgari et al. (2025). “Predicting the understandability of computational notebooks through code metrics analysis”. In: *Empirical Software Engineering* 30.4, p. 98.

Naziri, Amirreza, Arash Asgari, Aijun An, et al. (2025). “From Bias to Breakdown: Benchmarking Failure Mode Analysis of Single-cell RNA Sequencing Foundation Models in Acute Myeloid Leukemia”. In: *Proceedings of the AAAI Symposium Series*. Vol. 7. 1, pp. 553–557.

Naziri, Amirreza, Arash Asgari, Eleftherios Sachlos, et al. (2025). “Improving Classification of Cell Types in Acute Myeloid Leukemia with Self-guided Masking Technique”. In: *NeurIPS 2025 Workshop on AI Virtual Cells and Instruments: A New Era in Drug Discovery and Development*. NeurIPS, AI4D3.

Mojtaba Mostafavi Ghahfarokhi, Arash Asgari et al. (2024). “DistilKaggle: a distilled dataset of Kaggle Jupyter notebooks”. In: *Proceedings of the 21st International Conference on Mining Software Repositories*, pp. 647–651.

PROJECTS

ParsSum: Transformer-Based Persian Summarizers

Fine-tuned four near-SOTA Persian summarization transformers, curated two novel datasets, and deployed the end-to-end pipeline via a Django web app.

Persian News Sentiment Analysis Pipeline

Developed a custom NLP sentiment classification pipeline for news stories. Engineered and evaluated two distinct architectures: a fine-tuned Persian BERT and a custom-built RNN.

Additional Projects

Explore additional natural language processing and computer vision projects, including LLM fairness evaluation pipelines, dataset curation (DistilKaggle), and medical image segmentation.

[Source Code on GitHub](#)

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